

Social Media Engagement Analysis

One-page summary | 5,000 posts | Jan 2022 – 2023 | 10 countries, 4 post types, 8 content categories

5,000 posts analysed	10.1K avg likes / post	50.0K avg impressions	0.96% mean engagement rate	0.25% median engagement rate
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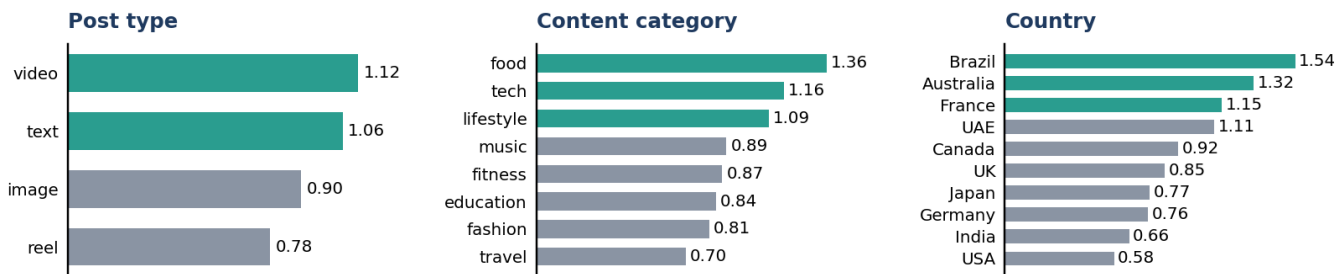
Objective & Data

The notebook explores what drives engagement on a social media dataset of 5,000 posts (19 fields covering user demographics, post attributes, interaction counts, reach and sentiment). Cleaning: no duplicates; 150 missing values each in age, gender, likes, comments, shares and sentiment were imputed (median / mode / median by post type; sentiment labelled “unknown”). Dates were parsed, a hashtag count and region mapping were derived, then statistical summaries, correlations and visualisations were produced.

Key Findings

- **Video and text lead on engagement rate** (1.12% and 1.06%); reels trail at 0.78%. Average likes and impressions are nearly identical across all four formats (~10.0–10.2K likes).
- **Food (1.36%), tech (1.16%) and lifestyle (1.09%)** are the strongest categories; travel is weakest (0.70%).
- **Brazil (1.54%) and Australia (1.32%)** top the country ranking; USA (0.58%) and India (0.66%) are lowest — even though India earns the highest average impressions (52.5K).
- **Verified accounts** see a modest edge in engagement (1.05% vs 0.95%) and watch time (4,101s vs 4,005s). **Mobile** users show the highest watch time (4,088s) and engagement (1.12%).
- **Sentiment** has little effect: negative posts average 1.04% engagement vs 0.95% for positive. **Age** is unrelated to engagement ($r = 0.01$).
- **Correlations are weak overall.** The only notable ones are impressions vs engagement rate ($r = -0.23$, since the rate is relative to reach) and likes vs engagement rate ($r = 0.09$).

Average engagement rate (%)



Caveats

Engagement rate is extremely right-skewed (max 191.5%, skewness 18.8, mean 0.96% vs median 0.25%), so group averages are heavily influenced by a few outliers and the gaps between groups should be treated as directional, not conclusive. Likes, comments, shares, watch time and impressions are close to uniformly distributed with near-zero correlation, which suggests the data may be synthetic. Time-of-day analysis was not possible because the data holds dates only, not posting times.

Recommendations

- Prioritise **video and text** formats and **food / tech / lifestyle** themes; review the reel strategy, which underperforms on engagement rate.
- Test content in **Brazil, Australia and France**; investigate why high-reach markets (India, USA) convert impressions to engagement poorly.
- Optimise for **mobile** viewing, and re-run the comparison using **median engagement rate** or outlier-trimmed data, ideally with posting timestamps added.